



Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  void function(int n) {
4      int count = 0;
5      int c = 0;
6      count++; // 1 step for the initialization statement: int c = 0;
7      count++; // For the initial/final False evaluation of the condition (i < n)
8      for (int i = n / 2; i < n; i++) {
9          count++; // For each successful True evaluation of (i < n)
10         count++; // For the initial/final False evaluation of the condition (j < n)
11         for (int j = 1; j < n; j = 2 * j) {
12             count++; // For each successful True evaluation of (j < n)
13             count++; // For the initial/final False evaluation of the condition (k < n)
14             for (int k = 1; k < n; k = k * 2) {
15                 count++; // For each successful True evaluation of (k < n)
16                 count++; // 1 step for the execution of the loop body statement (c++)
17                 c++;
18             }
19         }
20     }
21     printf("%d", count);
22 }
23
24 int main() {
25     int n;
26     scanf("%d", &n);
27     function(n);
28     return 0;
29 }
```

Check

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

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